The order of saddle points in tree tensor network training

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Abstract

We study gradient flow on tree tensor networks (TTNs) fitted to a target that is a sum of rank-one tensors. From small initialization the flow learns the rank-one terms one at a time and stays near a saddle point between consecutive terms. How long it stays is controlled by the order of the saddle, the lowest power of the distance at which the loss can decrease. We compute this order exactly at the saddle reached after one term has been learned. Typically it equals the number of cores in the smallest subtree that connects the cores on which some unlearned term differs from the learned one. This number depends on the shape of the tree, so the same target produces plateaus of different lengths on different trees. The order is lower only when the unlearned terms are, in a precise spectral sense, stronger than the learned one, and we give its value in that case too. At leading order, a saddle of order p is left after a time $\propto \ln(1/\sigma)$ if $p = 2$ and $\propto \sigma^{-(p-2)}$ if $p \geq 3$, where σ is the initialization scale; simulations on nine trees agree.

1 Introduction

In a deep linear network (DLN) trained from small initialization, gradient flow learns the singular modes of the target one at a time, with a plateau before each (Saxe et al., 2014; Jacot et al., 2022). Every plateau lasts $\propto \sigma^{-(K-2)}$, where K is the depth and σ the initialization scale. The reason is that the output is a product of K layers that all start at size σ: their common size a obeys $\dot{a} \approx c\, a^{K-1}$, whose solution from $a(0) = \sigma$ stays small for a time $\propto \sigma^{-(K-2)}$ and then grows quickly. In a DLN every mode needs every layer, so every plateau has the same exponent.

A TTN is also a product of K factors, its cores, but each core is attached to only some of the inputs. After the first term is learned, most cores are large. To add the next term, only some cores need to change from small values, and the length of the next plateau depends on how many. In the language of the loss landscape, it depends on the order of the saddle at which the flow rests.

This note determines that order (Theorem 1) and derives the resulting plateau lengths (Proposition 1). Section 6 compares the result with prior work.

2 Setting and definitions

Tree tensor networks. Let T be a tree with K nodes. Each node v holds a tensor G_v, called a *core*. A core has one leg for each tree edge at v and, possibly, one *physical* leg of dimension D_v. The legs on a tree edge e have dimension χ_e, the *bond dimension*. Contracting all tree-edge legs gives a tensor $W(\theta) \in \bigotimes_v \mathbb{R}^{D_v}$, where θ denotes all core entries. The network has N inputs $x_0, \ldots, x_{N-1} \in \mathbb{R}^d$, called *sites*; each site belongs to the physical leg of one core, so $D_v = d^{n_v}$ if core v carries n_v sites ($D_v = 1$ if it carries none). The network computes $f(x) = \langle W, x_0 \otimes \cdots \otimes x_{N-1} \rangle$.

Three trees serve as examples (Figure 1): the matrix product state (MPS), a path with one site per core; the binary tree, with two sites on each bottom core and none on the others; and the hierarchical Tucker (HT) tree, with one site on each leaf and none on the internal cores.

Target and loss. Labels are generated by a target tensor W^* as $y = \langle W^*, x_0 \otimes \cdots \otimes x_{N-1} \rangle$. For whitened Gaussian inputs the population squared loss is, up to a constant,

$$L(\theta) = \tfrac{1}{2} \|W(\theta) - W^*\|^2. \tag{1}$$

We take W^* to be a sum of distinct products of standard basis vectors,

$$W^* = \sum_r s_r \, U_r, \qquad U_r = e_{\pi_r(0)} \otimes e_{\pi_r(1)} \otimes \cdots \otimes e_{\pi_r(N-1)}, \qquad s_r > 0, \tag{2}$$

where π_r is a string of basis indices, one per site (for example $\pi_r = 110010$). We call U_r the *components* of the target. Distinct components are orthonormal.

The saddle after one component is learned. Fix a component j and let $q_v \in \mathbb{R}^{D_v}$ be the basis vector that U_j has on the sites of core v, so $U_j = \bigotimes_v q_v$. The point θ_S has cores

$$G_v = a_v \, q_v \otimes e_0 \otimes \cdots \otimes e_0, \qquad \prod_v a_v = s_j, \tag{3}$$

where e_0 is the first basis vector of each tree-edge leg. Then $W(\theta_S) = s_j U_j$: the network represents component j exactly, using one channel (index 0) on every edge. Every critical point with $W = s_j U_j$ has this form after a change of basis on the edges. The other $\chi_e - 1$ indices of each edge are *unused channels*. The loss at θ_S is $L_S = \frac{1}{2}\|\mathcal{R}\|^2$, with residual $\mathcal{R} = W^* - s_j U_j = \sum_{r \neq j} s_r U_r$.

Order of a saddle. The order of a critical point θ_S is the largest p such that, for some $C > 0$,

$$L(\theta) - L_S \;\geq\; -C \, \|\theta - \theta_S\|^p \qquad \text{for all } \theta \text{ near } \theta_S. \tag{4}$$

Equivalently, it is the smallest p for which some curve with $\|\theta(t) - \theta_S\| \leq t$ has $L(\theta(t)) - L_S \leq -ct^p$ (L is a polynomial, so the curve selection lemma applies). A saddle has order 2 iff its Hessian has a negative eigenvalue (a *strict* saddle). A saddle of order $p \geq 3$ has a positive semidefinite Hessian and the loss decreases only at order p.

Steiner subtrees. For a set B of cores, the *Steiner subtree* $\mathrm{St}(B)$ is the smallest subtree of T that contains B, and $|\mathrm{St}(B)|$ is its number of cores. For each component $r \neq j$ let

$$V_r = \{\text{cores carrying a site where } \pi_r \text{ and } \pi_j \text{ differ}\}, \qquad k_r = |\mathrm{St}(V_r)|, \qquad k^* = \min_{r \neq j} k_r. \tag{5}$$

Figure 1 gives an example. The point θ_S is a critical point iff $k^* \geq 2$ (Lemma 1).

3 Two ways to decrease the loss

Before stating the theorem we describe the two kinds of perturbation that lower the loss at θ_S. The theorem says that nothing else does better.

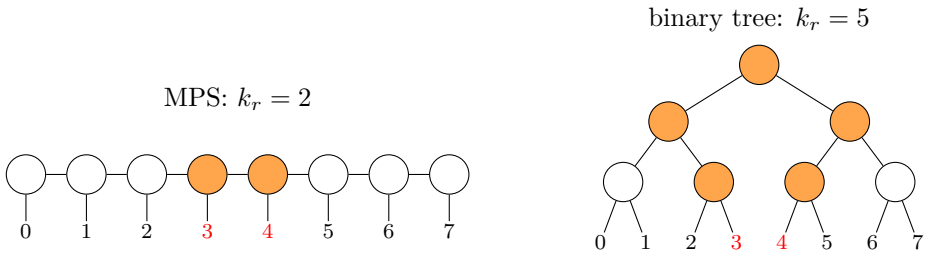


Figure 1: Learned component $\pi_j = 00000000$, unlearned $\pi_r = 00011000$. They differ at sites 3 and 4 (red). Orange: the Steiner subtree $\mathrm{St}(V_r)$. On the MPS the two sites are on adjacent cores ($k_r = 2$); on the binary tree the subtree passes through the root ($k_r = 5$); on the HT tree $k_r = 7$.

New channel along a subtree. An entry on an unused channel of edge e is multiplied, in the contraction, by the neighbouring core's entries on the same channel, which are zero at θ_S. So it changes W only if the neighbour also uses that channel. A set of cores therefore changes W through unused channels only if it is connected and all its cores use them together; the change is then a product of one small entry per core. To produce component r in this way, the set must contain every core on which π_r differs from π_j, hence all of $\mathrm{St}(V_r)$. With entries of size t on the k_r cores of $\mathrm{St}(V_r)$,

$$L - L_S = -c_r\, t^{k_r} + O(t^{2k_r}), \qquad c_r = s_r\, s_j^{(K-k_r)/K} \tag{6}$$

(Proposition 2, at $a_v = s_j^{1/K}$). So the order is at most k^*.

Combining two cheaper perturbations. A perturbation can also change the indices of W on a set B of cores at lower cost, in two ways:

- if $B = \{v\}$ is a single core, by rotating its vector q_v toward another basis vector, keeping the used channel; the change in W is linear in the rotation angle;
- if $|B| \geq 2$, by a new channel along $\mathrm{St}(B)$ as above; the change in W has size ϵ when each of the $|\mathrm{St}(B)|$ entries has size $\epsilon^{1/|\mathrm{St}(B)|}$.

Each such perturbation moves W away from $s_j U_j$, which costs $\frac{1}{2}s_j^2\epsilon^2$ at second order. Two of them on disjoint core sets B_1, B_2 also produce, at second order, the product tensor with modified indices on $B_1 \cup B_2$. If that tensor is an unlearned component r, this gains $s_r s_j \epsilon_1 \epsilon_2$. The net change is a quadratic form in the amplitudes, which is negative in some direction iff the gains beat the costs. The matrix below records the gains.

Definition 1 (Coupling matrix)**.** For $\ell \geq 1$, let $\mathcal{N}_\ell$ be the set of pairs (B, β), where B is a nonempty set of cores with $|\mathrm{St}(B)| \leq \ell$ and β assigns a basis index to each site of B such that, on every core of B, the indices differ from those of π_j. The coupling matrix M_ℓ is the symmetric matrix indexed by $\mathcal{N}_\ell$ with entries

$$M_\ell\big[(B_1,\beta_1),(B_2,\beta_2)\big] = \begin{cases} s_r & \text{if } B_1 \cap B_2 = \emptyset \text{ and the string equal to } \beta_1 \text{ on } B_1,\ \beta_2 \text{ on } B_2, \\ & \text{and } \pi_j \text{ elsewhere is } \pi_r \text{ for some component } r, \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

Let μ_ℓ be its largest eigenvalue, and

$$\ell^* = \min\{\ell < k^*/2 :\ \mu_\ell > s_j\} \qquad (\ell^* = \infty \text{ if there is no such } \ell). \tag{8}$$

M_ℓ has nonnegative entries, and μ_ℓ is nondecreasing in ℓ because M_ℓ is a principal submatrix of $M_{\ell+1}$. A nonzero entry requires a component r with $V_r = B_1 \sqcup B_2$, so if no component can be split into two pieces with Steiner subtrees of at most ℓ cores, then $M_\ell = 0$.

At level $\ell = 1$ only single-core rotations occur. The corresponding block of the Hessian at θ_S (balanced, $a_v = s_j^{1/K}$) is

$$H_{\mathrm{rot}} = s_j^{(K-2)/K}(s_j I - M_1), \tag{9}$$

so $\mu_1 > s_j$ iff this block has a negative eigenvalue. For $\ell \geq 2$ the combined perturbation has $\|\theta - \theta_S\| \sim \epsilon^{1/\ell}$ and lowers the loss by $\sim \epsilon^2$, which is order 2ℓ in the distance.

4 Main result

Theorem 1 (Order of the saddle after one learned component)**.** *Let W^* be as in (2) and θ_S as in (3), with $k^* \geq 2$. Assume*

(a) $\mu_\ell \neq s_j$ for every $\ell < k^*/2$;

(b) $\chi_e \geq 2$ on the edges of $\mathrm{St}(V_r)$ for some r with $k_r = k^*$; and, if $\ell^* < \infty$, there is a set $X \subseteq \mathcal{N}_{\ell^*}$ such that the submatrix of M_{ℓ^*} on X has largest eigenvalue above s_j and every edge e has $\chi_e - 1 \geq \#\{(B,\beta) \in X : |B| \geq 2,\ e \text{ is an edge of } \mathrm{St}(B)\}$.

Then

$$\mathrm{ord}(\theta_S) = \begin{cases} k^* & \text{if } \mu_\ell < s_j \text{ for all } \ell < k^*/2, \\ 2\ell^* & \text{otherwise.} \end{cases}$$

Assumption (a) excludes boundary cases. Assumption (b) provides the unused channels that the descent perturbations of Section 3 need. The theorem follows from three bounds, proved in Section 7: ord $\leq k^*$ (Proposition 2), ord $\geq \min(k^*, 2\ell)$ whenever $\mu_{\ell-1} < s_j$ (Proposition 3), and ord $\leq 2\ell$ whenever $\mu_\ell > s_j$ and $\ell < k^*/2$ (Proposition 4). The lower bound holds for any residual $\mathcal{R}$, not only sums of basis products.

Corollary 1. *Under the assumptions of Theorem 1:*

(i) If $k^* = 2$, the saddle is strict.

(ii) If $k^* \in \{3, 4\}$, the order is k^* if $\mu_1 < s_j$ and 2 if $\mu_1 > s_j$. So the order is k^* unless the Hessian has a negative eigenvalue.

(iii) If $W^* = s_j U_j + s_r U_r$ with $s_r < s_j$ (the larger component was learned), the order is k_r.

(iv) If $W^* = s_j U_j + s_r U_r$ with $s_r > s_j$, let $\ell_0 = \min \max(|\mathrm{St}(B_1)|, |\mathrm{St}(B_2)|)$ over partitions $V_r = B_1 \sqcup B_2$. The order is $2\ell_0$ if $\ell_0 < k_r/2$, and k_r otherwise.

(v) The order is below k^* while the Hessian is positive semidefinite only if $k^* \geq 5$.

Part (iii) covers the typical situation in training, where components are learned in decreasing order of weight: then the order is k_r, the size of a Steiner subtree, which depends on the tree.

Example 1 (Order below k^* with a positive semidefinite Hessian)**.** MPS with six cores, $d = \chi = 2$, learned $\pi_j = 000000$ with $s_j = 1$, and two unlearned components $\pi_1 = 110001$ and $\pi_2 = 110010$ with weight 0.8 each. Here $V_1 = \{0, 1, 5\}$, so $k_1 = 6$, and $V_2 = \{0, 1, 4\}$, so $k_2 = 5$; hence $k^* = 5$. No component differs from π_j on exactly two cores, so $M_1 = 0$ and the Hessian is positive semidefinite.

In M_2, the pair $(\{0,1\},11)$ couples with $(\{5\},1)$ through component 1 and with $(\{4\},1)$ through component 2, so $\mu_2 \geq 0.8\sqrt{2} \approx 1.13 > s_j$. Thus $\ell^* = 2 < k^*/2$ and the order is 4. The descent uses a new channel on cores $0,1$ that produces indices 11 there (the string 110000, not a component of W^*), together with rotations of cores 4 and 5. Numerically, $\min_{\|\theta-\theta_S\|=t} L - L_S$ has slope 4.000 in t and coefficient -0.0350, equal to the prediction $-(2\cdot 0.8^2-1)/8$. With weights 0.6, $\mu_2 = 0.85 < 1$ and the order is 5.

5 Plateau lengths

Proposition 1 (Escape time; derived at leading order)**.** *Start gradient flow at θ_S plus a perturbation of size ε along an order-p descent direction of Section 3. At leading order in ε, with the other variables relaxed to their instantaneous minimum, the amplitude b of the descent obeys $\dot{b} = c\, b^{p-1}$ with $c > 0$, so the time to leave the saddle is*

$$t_{\mathrm{esc}} \simeq \begin{cases} \ln(1/\varepsilon)/c, & p = 2, \\ \varepsilon^{-(p-2)}/((p-2)\,c), & p \geq 3. \end{cases} \tag{10}$$

For a new channel along $\mathrm{St}(V_r)$ (equal entries), $p = k_r$ and $c = c_r = s_r s_j^{(K-k_r)/K}$.

In a run from small initialization of scale σ, the entries that will grow are of order σ when the flow reaches θ_S, so the plateau there lasts $\propto \ln(1/\sigma)$ or $\propto \sigma^{-(p-2)}$. These entries already grow somewhat while component j is being learned. For $W^* = s_j U_j + s_r U_r$ with $s_r < s_j$, accounting for this growth gives the time Δt between learning j and learning r:

$$\Delta t \simeq \frac{1 - s_r/s_j}{s_r s_j^{(K-2)/K}} \ln\frac{1}{\sigma} \quad (k_r = 2), \qquad \Delta t \simeq \frac{1 - s_r/s_j}{(k_r-2)\, s_r s_j^{(K-k_r)/K}}\, \sigma^{-(k_r-2)} \quad (k_r \geq 3), \tag{11}$$

where the second formula assumes an initialization aligned with the target's basis vectors.

Evidence.

- *Order.* Minimizing $L - L_S$ numerically over spheres of radius t in 32 cases (MPS, binary and HT trees, k^* from 2 to 7, including residuals that are not sums of basis products) gives slopes within 0.003 of the order predicted by Theorem 1 in all 30 cases it covers; 12 further cases computed with independent code agree to $5 \cdot 10^{-4}$.
- *Two components, $s_r < s_j$.* In about 11,600 gradient-flow runs on nine trees (K from 3 to 15), the fitted exponent of Δt in $1/\sigma$ is $k_r - 2$ for $k_r = 3, 4, 5$, and for $k_r = 6, 7$ on two trees with balanced initialization. For $k_r = 2$ a logarithmic law fits better than a power law in all 14 settings, with slope 0.98–1.01 against the value 1 predicted by (11). On every edge outside $\mathrm{St}(V_r)$ the second singular value of the corresponding matricization of W stays below $4.3\,\sigma^2 s_r$.
- *Order $2\ell^* < k^*$.* A pre-registered test (about 2,800 runs) started gradient flow at θ_S plus a perturbation of size ε and fitted $t_{\mathrm{esc}} \propto \varepsilon^{-\gamma}$. For ε below the scale at which the order-$2\ell^*$ descent is predicted to overtake the order-k^* one, it found $\gamma = 2.001 \pm 0.001$ for $p = 4$ ($k^* = 5$ and 6) and $\gamma = 3.996$–3.998 for $p = 6$ ($k^* = 7$); the exponent $k^* - 2$ fits 10^3–10^5 times worse. Above that scale the results were mixed. This test has not been independently re-checked.

6 Relation to prior work

Deep linear networks. Saddle-to-saddle learning, the exponent $K-2$ and learning in order of decreasing singular value are established for DLNs (Saxe et al., 2014; Jacot et al., 2022). Achour et al. (2024, Thm 7) show which DLN saddles are strict, a lower bound of the form "order ≥ 3". A binary tree with four sites is exactly a depth-3 DLN; there $k^* = 3$ and Theorem 1 reduces to that classification.

Homogeneous models. Near the origin the weights of a homogeneous model align with a critical direction of the correlation with the target and escape after a time $\propto \sigma^{-(K-2)}$ (Kumar and Haupt, 2025). If 1, 2 or $k \geq 3$ layers start small while the others are large, the escape takes time $O(1)$, $\propto \ln(1/\sigma)$ or $\propto \sigma^{-(k-2)}$ (Rawal and DeWeese, 2026, Thm 4.2). Small weights keep the growth they acquired during earlier plateaus (Kunin et al., 2025; Zhang et al., 2026), which is the factor $1 - s_r/s_j$ in (11). These results take the number of small factors as given; Theorem 1 determines it for TTNs.

The landscape of TTNs. Furman et al. (2026) analyse the landscape of the same model and loss. They prove that local minima of minimum norm are global. At a minimum-norm critical point they consider the edges on which the rank of the model is below that of the target. Each connected set of such edges, touching n cores, gives a straight-line perturbation along which the loss decreases at order n (their Lemmas D.8–D.9), an upper bound on the order. They remark that with several separate such sets the smallest gives the fastest escape. They prove a lower bound for one saddle (a parity target, their Prop. 5.3) and do not study the dynamics.

What is new.

(1) A lower bound on the order at every saddle with one learned component, matching the upper bound, so the order is determined exactly. Previously lower bounds were known only for DLNs and one parity saddle.

(2) The order is set by the smallest single Steiner subtree $\mathrm{St}(V_r)$. For two components, the edges where the model rank is below the target rank are exactly the edges of $\mathrm{St}(V_r)$, so the upper bound of Furman et al. (2026) already equals k_r and the new part is the lower bound. With more components, those edges form the union of all the Steiner subtrees, often the whole tree, while the order is set by the smallest one.

(3) The case $\mu_\ell > s_j$: the order $2\ell^*$, and descent through index strings that are not components of the target.

(4) The escape laws (10)–(11) for TTNs and their numerical tests.

7 Proofs

Write $\varepsilon = \|\theta - \theta_S\|$, and let ν_v be the vector of entries of core v on unused channels (entries with at least one nonzero edge index).

7.1 Decomposition by the cores on which a tensor differs from U_j

Let $P_v = q_v q_v^\top$, $P_v^\perp = I - P_v$, and for a set B of cores

$$\Pi_B = \bigotimes_{v \in B} P_v^\perp \otimes \bigotimes_{v \notin B} P_v.$$

The 2^K projections Π_B are orthogonal and sum to the identity. Let H_B be the range of Π_B: the tensors that differ from U_j exactly on the cores of B. A basis of H_B is given by the products of basis vectors whose indices differ from those of π_j on every core of B, and $H_{B_1} \otimes H_{B_2} \cong H_{B_1 \sqcup B_2}$ for disjoint sets. Let $\mathcal{R}_B = \Pi_B \mathcal{R}$. By (2),

$$\mathcal{R}_B = \sum_{r:\, V_r = B} s_r U_r, \qquad \text{so} \qquad k^* = \min\{|\mathrm{St}(B)| : \mathcal{R}_B \neq 0\}. \tag{12}$$

Lemma 1. *(i) θ_S is critical iff $\mathcal{R}_{\{v\}} = 0$ for every core v, i.e. iff $k^* \geq 2$. (ii) Let $N(\theta) = \langle U_j, W(\theta)\rangle$ and, for $B \neq \emptyset$, $m_B = \Pi_B W / N$. Wherever $N \neq 0$,*

$$L(\theta) - L_S = \tfrac{1}{2}(N - s_j)^2 + \sum_{B \neq \emptyset} \Big[\tfrac{N^2}{2}\|m_B\|^2 - N\langle \mathcal{R}_B, m_B\rangle\Big]. \tag{13}$$

Proof. (i) Entries on unused channels do not change W at first order (Section 3). A change of the used entries of core v changes W within $H_\emptyset \oplus H_{\{v\}}$. So the gradient vanishes iff $\mathcal{R}$ has no component in any $H_{\{v\}}$; its component in $H_\emptyset$ is zero because $\langle U_j, \mathcal{R}\rangle = 0$. (ii) By orthogonality, $\|W - W^*\|^2 = \sum_B \|\Pi_B W - \Pi_B W^*\|^2$, with $\Pi_\emptyset(W - W^*) = (N - s_j)U_j$, and for $B \neq \emptyset$, $\Pi_B W^* = \mathcal{R}_B$ and $\Pi_B W = N m_B$. Also $L_S = \frac{1}{2}\sum_B \|\mathcal{R}_B\|^2$. □

7.2 Upper bound k^*

Proposition 2. *Let $k_r = k^*$ and $\chi_e \geq 2$ on the edges of $\mathrm{St}(V_r)$. At $a_v = s_j^{1/K}$, set the entry of each core $v \in \mathrm{St}(V_r)$ with physical index given by π_r on the sites of v, index 1 on the edges of $\mathrm{St}(V_r)$ and index 0 on its other edges, to t. Then $\|\theta(t) - \theta_S\| = \sqrt{k^*}\, t$ and*

$$L(\theta(t)) - L_S = -s_r\, s_j^{(K-k^*)/K}\, t^{k^*} + \tfrac{1}{2} s_j^{2(K-k^*)/K}\, t^{2k^*}.$$

Proof. Expand the contraction by choosing, for each core of $\mathrm{St}(V_r)$, either its original entry or the new one. If a term uses the new entry on one core of $\mathrm{St}(V_r)$ and the original entry on a neighbouring core of $\mathrm{St}(V_r)$, the shared edge contracts index 1 with index 0 and the term vanishes. Only two terms remain: all original entries, giving $s_j U_j$, and all new entries on $\mathrm{St}(V_r)$, giving $t^{k^*} s_j^{(K-k^*)/K} U$, where U has the indices of π_r on the sites of $\mathrm{St}(V_r)$ and those of π_j elsewhere. Since $\pi_r = \pi_j$ outside V_r, $U = U_r$. Hence $W - W^* = -\mathcal{R} + t^{k^*} s_j^{(K-k^*)/K} U_r$, and $\langle \mathcal{R}, U_r\rangle = s_r$. □

7.3 Lower bound

Cumulants. Consider families $f = (f_B)_{B \subseteq V}$ with $f_B \in H_B$, with the product $(f * g)_B = \sum_{B_1 \sqcup B_2 = B} f_{B_1} \otimes g_{B_2}$. This product is commutative and associative with unit $(1, 0, 0, \dots)$, and every family with $f_\emptyset = 0$ is nilpotent, so exp and log are finite sums and $\log(f * g) = \log f + \log g$. With $m_\emptyset = 1$, define $\kappa = \log m$. Explicitly,

$$m_B = \sum_{\Pi \in \mathrm{Part}(B)} \bigotimes_{C \in \Pi} \kappa_C, \qquad \kappa_B = \sum_{\Pi \in \mathrm{Part}(B)} (-1)^{|\Pi|-1}(|\Pi| - 1)! \bigotimes_{C \in \Pi} m_C,$$

sums over set partitions of B. The m_B and κ_B are analytic near θ_S and vanish there for $B \neq \emptyset$, so $\|\kappa_B\| \leq C_0\varepsilon$.

Lemma 2. *For $|B| \geq 2$, near θ_S, $\|\kappa_B\| \leq C_0 \prod_{x\in \mathrm{St}(B)} \|\nu_x\|$. In particular $\|\kappa_B\| \leq C_0\varepsilon^{|\mathrm{St}(B)|}$.*

Proof. Fix $x \in \mathrm{St}(B)$ and suppose $\nu_x = 0$. Then all nonzero entries of G_x have edge indices 0, so W factorizes at x: $W = g_x \otimes \bigotimes_b W_b$, where g_x is the physical vector of G_x, b runs over the components of T with x removed (node sets V_b), and W_b is the contraction of component b with its edge to x set to 0. The projections Π_B factorize core by core, so $m = m^{(x)} * \prod_b m^{(b)}$ with $m^{(x)}$ supported on $\{\emptyset, \{x\}\}$ and $m^{(b)}$ on subsets of V_b. Hence $\kappa = \log m^{(x)} + \sum_b \log m^{(b)}$ is supported on subsets of $\{x\}$ or of a single V_b. B is neither: if $x \in B$, then $B \not\subseteq \{x\}$ and $B \not\subseteq V_b$; if $x \notin B$ and $B \subseteq V_b$, then the connected set V_b contains $\mathrm{St}(B)$, which contains x. So $\kappa_B = 0$ whenever $\nu_x = 0$ for some $x \in \mathrm{St}(B)$. In the power series of κ_B around θ_S, every monomial therefore contains a variable of ν_x for each $x \in \mathrm{St}(B)$; factoring these out and bounding the rest on a small polydisc gives the claim. □

Proposition 3. *Let θ_S be critical and $2 \leq \ell \leq \lceil k^*/2 \rceil$. Let $S_{\ell-1}$ be the symmetric operator on $\bigoplus_{|\mathrm{St}(C)|\leq \ell-1} H_C$ with blocks $(x, y) \mapsto \langle \mathcal{R}_{C_1\sqcup C_2}, x \otimes y\rangle$ for disjoint C_1, C_2 and zero otherwise. If $\lambda_{\max}(S_{\ell-1}) < s_j$, then $L(\theta) - L_S \geq -C\varepsilon^{\min(k^*,2\ell)}$ near θ_S. In the basis of products of basis vectors, $S_{\ell-1} = M_{\ell-1}$.*

Proof. Let $p = \min(k^*, 2\ell)$ and $y_C = \|\kappa_C\|$. Call C *small* if $|\mathrm{St}(C)| \leq \ell - 1$ and *large* otherwise. Then: (F1) a large set has $|\mathrm{St}(C)| \geq \ell \geq 2$, so $y_C \leq C_0\varepsilon^\ell$ by Lemma 2; (F2) subsets of small sets are small; (F3) $\mathcal{R}_C = 0$ for small C, since $|\mathrm{St}(C)| < k^*$; (F4) summing $\langle \mathcal{R}_B, \kappa_{C_1} \otimes \kappa_{C_2}\rangle$ over all B and all unordered partitions $B = C_1 \sqcup C_2$ into small sets gives $\frac{1}{2}\langle \kappa_{\mathrm{s}}, S_{\ell-1}\kappa_{\mathrm{s}}\rangle \leq \frac{1}{2}\lambda_{\max}(S_{\ell-1})\|\kappa_{\mathrm{s}}\|^2$, where κ_{s} collects the cumulants of small sets.

Quadratic term of (13). For small B, $m_B - \kappa_B$ is a sum of products of at least two cumulants of small sets (F2), each at most $C_0\varepsilon\, y_C$ for one factor C. So $\sum_{B \text{ small}} \|m_B\|^2 \geq (1 - C_2\varepsilon)\|\kappa_{\mathrm{s}}\|^2$; the terms with large B are nonnegative and dropped.

Linear term. Only B with $\mathcal{R}_B \neq 0$ contribute, and they have $|\mathrm{St}(B)| \geq k^*$. Expanding m_B over set partitions: the single cumulant gives $O(\varepsilon^{k^*})$ (Lemma 2); two small blocks give (F4); one small and one large block give at most $\frac{\gamma}{2}y_{C_1}^2 + C_\gamma\varepsilon^{2\ell}$ (Young's inequality and F1); two large blocks give $O(\varepsilon^{2\ell})$; three or more blocks give $O(\varepsilon)(y_{C_1}^2 + y_{C_2}^2)$ if two blocks are small and $O(\varepsilon^{2\ell})$ otherwise. Summing, $\sum_B \langle \mathcal{R}_B, m_B\rangle \leq \frac{1}{2}(\lambda_{\max}(S_{\ell-1}) + M\gamma + C\varepsilon)\|\kappa_{\mathrm{s}}\|^2 + C_\gamma\varepsilon^p$.

Conclusion. By (13), dropping $\frac{1}{2}(N - s_j)^2$, $L - L_S \geq \frac{N}{2}[N(1 - C_2\varepsilon) - \lambda_{\max}(S_{\ell-1}) - M\gamma - C\varepsilon]\|\kappa_{\mathrm{s}}\|^2 - NC_\gamma\varepsilon^p$. With $g = s_j - \lambda_{\max}(S_{\ell-1}) > 0$, $\gamma = g/(4M)$ and $N \geq s_j - C\varepsilon$, the bracket is positive for small ε, so $L - L_S \geq -C\varepsilon^p$. □

7.4 Upper bound 2ℓ

Proposition 4. *Let $\ell < k^*/2$ and let $X \subseteq \mathcal{N}_\ell$ satisfy assumption (b) of Theorem 1 with ℓ in place of ℓ^*, with largest eigenvalue $\rho > s_j$ of M_ℓ restricted to X. Then some curve has $\|\theta(t) - \theta_S\| \leq t$ and $L(\theta(t)) - L_S \leq -ct^{2\ell}$.*

Proof. Take $a_v = a = s_j^{1/K}$ and a unit eigenvector $x \geq 0$ of $M_\ell|_X$ for ρ (Perron–Frobenius). For $0 < \eta \leq 1$:

- for $(\{v\},\beta) \in X$, set the entry of G_v with physical index β and all edge indices 0 to $a\eta x_{(\{v\},\beta)}$ (a rotation of q_v);
- for $n = (B,\beta) \in X$ with $|B| \geq 2$, let $h = |\mathrm{St}(B)| \leq \ell$. On every edge of $\mathrm{St}(B)$ choose an unused index not chosen by any other such element of X (possible by (b)). Give each core $u \in \mathrm{St}(B)$ the entry $a(\eta x_n)^{1/h}$ with physical index β (if $u \in B$) or that of π_j (if $u \notin B$), the chosen indices on the edges of $\mathrm{St}(B)$, and 0 on its other edges.

Then $\|\theta - \theta_S\| \leq C\eta^{1/\ell}$. In a nonzero term of the contraction, every connected set of edges carrying nonzero indices uses the entries of a single element n and equals its $\mathrm{St}(B)$. So W is a sum over collections of elements of X on disjoint cores, each contributing the factor ηx_n and changing the indices on its set B. In particular $N = s_j$, and $m_B = \eta m_B^{(1)} + \eta^2 m_B^{(2)} + O(\eta^3)$, where $m^{(1)}$ collects single elements and $m^{(2)}$ pairs.

Order η. $\sum_B \|m_B^{(1)}\|^2 = \|x\|^2 = 1$. No single element has $B = V_r$, since that would need $|\mathrm{St}(B)| \geq k^* > \ell$; so $\sum_B \langle \mathcal{R}_B, m_B^{(1)} \rangle = 0$ by (12).

Order η^2. If $M_\ell[n_1, n_2] = s_r > 0$, then $\mathrm{St}(B_1)$ and $\mathrm{St}(B_2)$ are disjoint: otherwise their union would be a connected set containing V_r with at most $2\ell - 1 < k^*$ cores. So every nonzero entry of $M_\ell|_X$ corresponds to a pair on disjoint cores, and $\sum_B \langle \mathcal{R}_B, m_B^{(2)} \rangle = \frac{1}{2} x^\top M_\ell|_X x = \frac{\rho}{2}$.

By (13) with $N = s_j$: $L - L_S = \frac{s_j^2}{2}\eta^2 - \frac{s_j \rho}{2}\eta^2 + O(\eta^3) = -\frac{s_j}{2}(\rho - s_j)\eta^2 + O(\eta^3)$. Set $\eta = (t/C)^\ell$. □

7.5 Proof of Theorem 1 and Corollary 1

If $k^* = 2$, the curve of Proposition 2 has order 2, and every critical point has order at least 2. Let $k^* \geq 3$.

All $\mu_\ell < s_j$. Proposition 3 with $\ell = \lceil k^*/2 \rceil$ (using $\mu_{\lceil k^*/2 \rceil - 1} < s_j$) gives ord $\geq k^*$, and Proposition 2 gives ord $\leq k^*$.

Some $\mu_\ell > s_j$. If $\ell^* = 1$, Proposition 4 gives ord ≤ 2, so ord $= 2$. If $\ell^* \geq 2$, then $\mu_{\ell^* - 1} < s_j$ by minimality and (a), so Proposition 3 at $\ell = \ell^*$ gives ord $\geq 2\ell^*$, and Proposition 4 gives ord $\leq 2\ell^*$. □

Corollary. (i) There is no $\ell < 1$. (ii) The only level below $k^*/2$ is $\ell = 1$. For (9): rotating q_u by ϵ_u toward another basis vector changes W by an orthonormal tensor times $a^{K-1}\epsilon_u$, and two rotations on cores u, v couple through any component with $V_r = \{u, v\}$ with coefficient $-s_r a^{K-2}$. So the block is $a^{2(K-1)} I - a^{K-2} M_1 = s_j^{(K-2)/K}(s_j I - M_1)$. (iii) With one unlearned component, each row of M_ℓ has at most one nonzero entry, s_r, so $\mu_\ell \leq s_r < s_j$. (iv) Same, with $\mu_\ell = s_r > s_j$ from the first ℓ that admits a partition, namely ℓ_0. (v) For $k^* \leq 4$, an order below k^* requires $\mu_1 > s_j$, which makes the Hessian indefinite. □

7.6 Escape time (Proposition 1)

New channel along $\mathrm{St}(V_r)$. With the other entries fixed at θ_S and amplitudes b_v on the cores of $\mathrm{St}(V_r)$, Proposition 2 gives $\dot{b}_v = A \prod_{u \neq v} b_u (s_r - A \prod_u b_u)$ with $A = a^{K - k_r}$. For equal small amplitudes, $\dot{b} = s_r A b^{k_r - 1}$, and $\frac{d}{dt} b^{-(p-2)} = -(p-2)c$ gives (10).

Combined perturbation. With one element of size $|B| \geq 2$ (as in Example 1), the rotations have $O(1)$ curvature and follow the amplitude $y = \prod_{v \in \mathrm{St}(B)} b_v$. Minimizing over them leaves $L - L_S \approx -\frac{c}{2} y^2$ with $c > 0$ (in Example 1, $c = 0.8^2 + 0.8^2 - 1$). Then $\dot{b}_v = c\, b_v \prod_{u \neq v} b_u^2$, and for equal amplitudes $\dot{b} = c b^{2\ell - 1}$.

Growth before the plateau. While j is learned from aligned initialization, its cores grow as $\dot{a} \simeq s_j a^{K-1}$ and the entries on $\mathrm{St}(V_r)$ already grow at rate $s_r a^{K-k_r} b^{k_r-1}$. For $k_r \geq 3$, $b^{-(k_r-2)}$ decreases by $(k_r-2)s_r \int a^{K-k_r} dt = (k_r-2)s_r \int a^{1-k_r} da/s_j = \frac{s_r}{s_j}\sigma^{-(k_r-2)} + O(1)$, so at the plateau $b^{-(k_r-2)} = (1 - s_r/s_j)\sigma^{-(k_r-2)} + O(1)$. For $k_r = 2$, $\ln b$ grows by $\frac{s_r}{s_j} \ln \frac{1}{\sigma} + O(1)$. Inserting into (10) gives (11).

8 Scope

The theorem covers saddles with one learned component, targets that are sums of products of basis vectors (the lower bound holds for any target), whitened inputs and the population loss. Saddles with several learned components and the boundary cases $\mu_\ell = s_j$ are open. Proposition 1 is a leading-order derivation, not a theorem. When $2\ell^* < k^*$ both kinds of perturbation are present, and the lower order wins only below a scale set by their coefficients.

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